# Problem Set 4

Version of October 21, 2026

**Problem 4.1** (One-stage cluster sampling). The California Academic Performance Index (API) was a measure of school performance based on standardized tests. API scores ranged from 200 to 1,000, with higher values indicating better performance. The last API report was produced in 2013.

The API data are included in the `survey` package and can be loaded with `data(api)`. We use the one-stage cluster sample `apiclus1`. A simple random sample of $n = 15$ school districts was drawn from $N = 757$ districts, and all schools in the selected districts were observed.

The survey design is specified as follows:

```
dn_api <- svydesign(id = ~dnum, weights = ~pw, fpc = ~fpc, data = apiclus1)
```

Here, `dnum` identifies the school district, `pw` contains the design weights, and `fpc` contains the finite-population correction for sampling districts. We focus on `api00`, the API score in 2000.

a) Estimate the population mean API score in 2000. Also compute the design effect by calling `svymean()` with `deff = TRUE`.

b) Estimate the 25th percentile, median, and 75th percentile of the API score in 2000. Use

   ```
   svyquantile(x, design, quantiles, ...)
   ```

c) The California State Board of Education used an API score of 800 as a performance target. Estimate the proportion of schools with an API score below 800. Use

   ```
   svycdf(formula, design, ...)
   ```

**Problem 4.2** (Domain estimation). We use the data set `sepe.dat` from Problem Set 1. Treat the data as a simple random sample without replacement (SRS) of $n = 800$ establishments from a population of size $N = 3770$. We focus on industry class 31, that is, the domain $U_{31}$ identified by `code == 1`.

a) Assume that the domain size $N_{31}$ is unknown. Estimate the domain population mean $\bar{y}_{U31}$ of environmental-protection expenditures and its standard error.

   **Hint.** If `dn` is a survey design created with `svydesign()`, use `subset(dn, code == 1)` to restrict the design to the domain. Do not create a new SRS design from the domain observations: under the original SRS, the domain sample size $n_{31}$ is random.

b) Now assume that the domain size $N_{31} = 650$ is known, but that this information was not used when the sample was drawn. Estimate the population domain total of environmental expenditures $t_{y,31}$ and its standard error.

**Problem 4.3** (Stratified simple random sampling). We again use the data set `env.dat`. Assume that membership in industry class 31 was known from the sampling frame and that the population was stratified by this variable before sampling. Within each stratum, a simple random sample without replacement was drawn. The population consists of two strata:

| Stratum | Population size |
|---|---|
| $U_{31}$ (`code == 1`) | $N_{31} = 460$ |
| $U_{\text{rest}}$ (`code == 0`) | $N_{\text{rest}} = 3{,}310$ |

**Hint.** The stratum sample sizes are available in `sepe.dat`. As explained in Problem Set 2, create a survey design object for stratified sampling and name it `dn_sepe`.

a) Estimate the population total of environmental-protection expenditures using `svytotal()`. Also report a 90 % confidence interval using

```
confint(object, parm, level = 0.95, ...)
```

b) Estimate the mean environmental-protection expenditures in each stratum using `svyby()`.

**Hint.** The function `svyby(formula, by, design, FUN, ...)` applies the survey statistic `FUN` to the variables specified by `formula`, separately for the groups defined by `by`, using the survey design `design`. Additional arguments can be passed through `...`.

Next, test whether the two stratum means are equal. For comparison, use both

```
t.test(expend ~ code, data = sepe, var.equal = TRUE)
```

and the design-based test

```
svyttest(expend ~ code, dn_sepe)
```

The first is the classical pooled two-sample $t$-test and ignores the sampling design. The second uses the variance implied by the survey design, including the finite population corrections.

c) Let $y_i$ denote environmental-protection expenditures (`expend`) and $x_i$ the number of full-time employees (`empl`). Two different summaries are commonly considered:

$$r_{AR} = \frac{1}{N_+} \sum_{i \in U_+} \frac{y_i}{x_i}, \qquad r_{RA} = \frac{\sum_{i \in U} y_i}{\sum_{i \in U} x_i},$$

where $U_+ = \{i \in U : x_i > 0\}$ and $N_+ = |U_+|$. The first is the average of ratios; the second is the ratio of totals (equivalently, the ratio of averages).

Estimate $r_{RA}$ using `svyratio()` and the design object `dn_sepe`. Then repeat the calculation with `separate = TRUE`. What changes?

**Hint.** Use

```
svyratio(numerator = formula, denominator, design, ...)
```

The default estimates one combined ratio. With `separate = TRUE`, ratios are estimated separately within the strata.

d) Convert `dn_sepe` to a replicate-weight design for the Rao–Wu $(n-1)$ bootstrap with $B = 200$ replicates:

```
set.seed(11)
dn_sepe_boot <- as.svrepdesign(dn_sepe, type = "subbootstrap",
                               replicates = 200)
```

Re-estimate the combined ratio $r_{RA}$ from part c) using this replicate design. Also report its 95 % confidence interval.